THE BASE CHANGE OF A DOMINANT MORPHISM ALONG AN OPEN MORPHISM IS DOMINANT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted statement carries the source label lemma-open-base-change-dominant; following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the label is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The counted spans contain no \ref, no \eqref and no citation, so no external reference stub and no bibliography excerpt are attached to this card; this was verified by scanning the two spans rather than by trusting the pool card alone.

Editorial note. The proof is the author's complete argument: it does not end in a pointer, an omission or a 'follows easily', and every step that the contradiction needs is present in the counted proof span.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of sources/stacks-chow/preamble.tex lines 14-18 and 154-217. No macro was invented or redefined, and this card adds no preamble text of its own.

Allowed standard prerequisites (editorial).

- A morphism of schemes is dominant if and only if its image is dense, so a morphism fails to be dominant exactly when its image is contained in a proper closed subset. Same chapter, the notion is used in the counted statement at line 1336 and in the counted proof at lines 1342-1343. Supplies the form of the hypothesis and the shape of the contradiction argument: not dominant means the image lies in a closed subset B different from S'.
- Fibre products of schemes and the universal property of the base change, in the form that points of a fibre product over a field-valued point are pairs of points with the same image. Used inside the counted proof at lines 1346-1350, where the base is changed to W = Spec(kappa(f(x))). Gives that X_W and (S' minus B)_W nonempty forces their fibre product over W to be nonempty, which is the contradiction that closes the proof.

Lemma 0.1. *Let $f : X \to S$ be a dominant morphism of schemes. Let $g : S' \to S$ be an open morphism of schemes, and let $f' : X' \to S'$ be the base change of f by g. Then f' is dominant.*

Proof. If f' is not dominant, then $f'(X')$ is contained in some closed subset B not equal to S'. Then $S' - B$ is open in S' and nonempty. Since g is open, $g(S' - B)$ is nonempty and open in S. Since $f(X)$ is dense in S, there is a point $x \in X$ with $f(x) \in g(S' - B)$. Let $W = \mathrm{Spec}(\kappa(f(x)))$, and write X_W, S_W, S'_W for the pullbacks of these schemes from S to W. Then X_W and $(S' - B)_W$ are nonempty,

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

and so $X_W \times_W (S' - B)_W$ is nonempty, as W is the spectrum of a field. This contradicts that $f'(X \times_S S')$ is contained in $B \subset S'$. So in fact f' is dominant. $\square$